

EE 618: Industrial Process Control

Course Code	EE 618
Course Name	Industrial Process Control
L-T-P-C	2.5-0.5-0-3
Prerequisites	EE 514 Linear Dynamical Systems or teachers consent
Intended for	U.G. and P.G. Distribution: Elective Semester: Even
Approval	9th Senate (May 2016)

Course Contents

- **Introductory concepts of Process control:** The Chemical Process, An Industrial Perspective of a Typical Process Control Problem, Variables of a Process, The Concept of a Process Control System, Introduction to Control System Implementation, Instrumentation, Material and Energy Balances, Form of Dynamic Models, Linear Models and Deviation Variables.[3 Lectures]
- **Process dynamics, modelling and identification:** Models for dynamical systems, Similarity transformations and minimal representations, Analysing linear dynamical systems, Development of Theoretical Process Models, Parameter Estimation, Validation of Theoretical Models, Principles of Empirical Modelling, Step-Impulse-Frequency response Identification, Closed-loop stability, Control performance in different frequency ranges, Unstable systems, Limitations due to uncertainty in the plant model and input constraints. [6 Lectures]
- **Single-loop and multivariable process control:** Control loop structures for the regulatory control layer, Feed-forward control, Ratio Control, Cascade control, Auctioneering control, split range control, parallel control, Controller tuning using Fundamental process models, approximate process models, frequency response models, tuning without a Model, Design of More Complex Control Structures, Controller Design for Processes with Difficult Dynamics, Controller Design for Nonlinear Systems, Nature of Multivariable Systems, Open-Loop and closed loop Dynamic Analysis, Relative Gain Array, Loop Pairing and shaping, Decoupling, Steady- State Decoupling by Singular Value analysis, Model-Based Controllers for Multivariable Processes, Tuning of decentralised controllers. [12 Lectures]
- **Control structure selection and plant-wide control:** Top-down analysis, bottom-up design, regulatory control, determining degrees of freedom, selection of controlled variables based on local analysis, selection of manipulated variables, mass balance control and throughput manipulation, economic considerations in plantwide control. [5 Lectures]
- **?? Model-based predictive control:**
- Formulation of a QP problem for MPC, step response models, updating the process model, Kalman filters, disturbance handling and off-set free control, feasibility and constraint handling, closed-loop stability with MPC controllers, target calculation, robustness of MPC controllers. [6 Lectures]
- **Special topics in process control:** Discrete time implementation - aliasing - sampling interval, pure integrators in parallel, anti-windup control, Hanuss self conditioned form, observer-based controllers, bumpless transfer, nonlinear systems - methods of dynamical analysis and linearization, basics of process monitoring and diagnosis, linear regression techniques applied in process control - principal component analysis - partial least squares - Fourier-Motzkin elimination. [6 Lectures]
- **Lab Exercises:** [4 lectures] Experiments on process control applications on water tank system, robotic arm, etc.

Text Books

1. B. A. Ogunnaike, W. H. Ray, **Process Dynamics, Modelling and Control**, Oxford University Press, 1994.
2. D. E. Seborg, T. E. Edgar, D. A. Mellichamp, **Process Dynamics and Control**, John Wiley and Sons, 2004.

References

1. B. W. Bequette, **Process Control: Modeling, Design, and Simulation**, Prentice Hall International Series, 2002.
2. F. G. Shinskey, **Process Systems: Application, Design and Adjustments**, McGraw Hill, 1967.